# Identifying Effective Municipal Education Management Practices in Brazil

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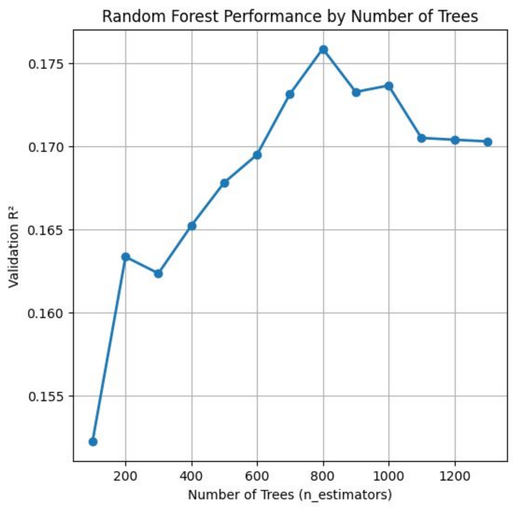
## Problem

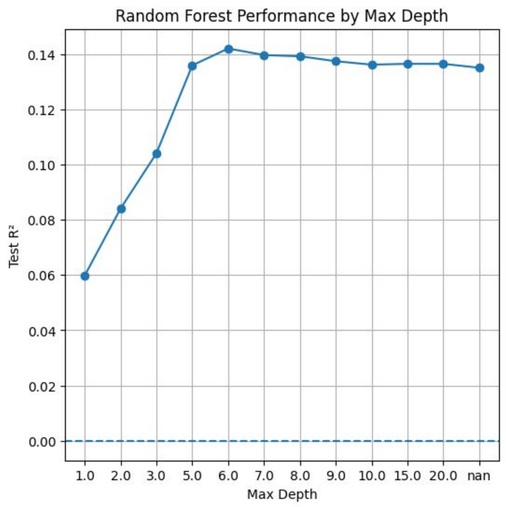
We aim to identify which practices of municipal education secretaries in Brazil influence student math performance in public schools, as measured by SAEB 2023 scores, a national exam assessing student learning outcomes. Inspired by prior work in Tanzania (Cilliers et al., 2022), we use a Random Forest regression model to identify the key predictors of math scores, using feature importance, correlation analysis, and sequential feature selection. These insights highlight some of the factors associated with student outcomes, while illustrating the challenges of understanding the drivers of educational performance in local government-run schools.

## Baseline model

The first Random Forest model was trained using all initial variables to establish a baseline performance and compute feature importance scores. These results were used to identify the most relevant variables and guide subsequent feature selection and model refinement.

## Hyperparameter tuning




## Data

The dataset is based on the SAEB (Basic Education Assessment System) questionnaire completed by municipal Secretaries of Education across Brazil. It captures information on governance practices such as teacher training, interactions with principals, and administrative processes.

Original dataset: 5,568 municipalities and 176 variables. We removed open-ended and non related questions, as well as variables and observations with substantial missing data.

Final dataset: 1,267 municipalities, 102 independent variables and the SAEB Math 2023 (municipal score).

## Random forest model

After the baseline model, we selected the top variables based on their correlation with the target and their feature importance scores. We then applied Sequential Feature Selection, which iteratively builds regression models by adding variables and evaluating their performance. At each step, the combination of variables that maximizes predictive accuracy ($R^2$) is retained. This process identifies the subset of features that provides the best-performing model while reducing dimensionality.

$$\hat{f}(x) = \frac{1}{B}\sum_{b=1}^{B} T_b(x)$$

$\hat{f}(x)$: predicted outcome (SAEB math score)
$B$: number of trees (n_estimators)
$T_b(x)$: prediction from tree $b$
$x$: vector of predictors (municipal practices)

## Results and future

Best predictors (selected variables): Secretary experience, Standardized curriculum developed by external organization, Budget decision autonomy, Existence of a municipal education system and an education forum, Special education organizations collaboration, Principal selection (management experience), Budget allocation privority given to school construction budget al priority and material purchase, SAEB score-based school rewards.

**Baseline model** → Training $R^2$: 0.87; Test $R^2$: 0.14 (high variance, overfitting)

**Otimized model** → Training $R^2$: 0.16; Test $R^2$: 0.08 (lower variance)

Low $R^2$ is expected given the complexity of education outcomes and limited power of survey-based variables. Future research could test different model specifications and more systematic hyperparameter tuning. Finally, validating the model across different years or datasets could help assess robustness and generalizability.

Cilliers, J., Dunford, E., & Habyarimana, J. (2022). What do local government education managers do to boost learning outcomes? https://doi.org/10.1596/41096
Dalmon, D. L. (2025). *An Analysis of the Influence of Municipal Governments' Practices on Primary Education Learning Achievement in Brazil*.